

ALU explorer

An arithmetic logic unit takes two operands, an opcode, and produces a result plus condition flags

ADD five plus three starter

- Starter: four-bit ADD, A equals five, B equals three
- Result Y is one-zero-zero-zero binary
- Flags: Z zero, N one, C zero, V zero
- Switch opcode to SUB, AND, or SLT and watch Y and flags change
- On ADD, C equals one means unsigned carry out
- On SUB, C equals one means no borrow

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

ALU operation explorer

Pick an opcode, set A and B, and watch the result plus flags **Z** (zero), **N** (MSB), **C** (carry/borrow), and **V** (signed overflow). Starter: 4-bit ADD of 5 + 3.

Starter example: 4-bit ADD, A=5, B=3 → $Y=1000_2$ (8), flags Z=0 N=1 C=0 V=0.

Challenge 0 / 22 cleared

ADD 5+3: 4-bit ADD with A=5, B=3 → Y=8, Z=0.

Show hintCheckNextLoad challenge setupIdle

ADD 5+3

ADD carry

ADD overflow V

SUB 9-4

SUB borrow

AND

OR

XOR

NOT

SHL

SHR

SLT unsigned

SLT signed

SLT vs SLT.S

Zero flag

Negative flag

8-bit ADD

Quiz: Z

Quiz: V

Quiz: C on ADD

Quiz: ALU role

NAND

ALU

Load starter example

WIDTH

A (DECIMAL)

B (DECIMAL)

4-bit

5

3

ADD

SUB

AND

OR

XOR

NOT

NAND

NOR

SHL

SHR

SLT

SLT.S

PASSA

PASSB

ALU explorer starter

Workbook practice

- In the workbook track, compute four-bit ADD five plus three and list Z, N, C, V
- Do SUB three minus five and explain C equals zero
- AND twelve with three for zero and Z equals one
- Explain when unsigned SLT and signed SLT.S disagree on fifteen versus one
- Name one pitfall: reading C as signed overflow instead of V

Pitfalls to watch

- Do not mix unsigned carry with signed overflow, C and V mean different things
- N is MSB of result, not a separate sign register
- Compare ops return zero or one in Y, not a full difference
- And remember: the browser lab is literacy
- Real ALUs still need decode, timing, and ISA-specific flag rules

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few challenges after the starter
- On paper, fill one opcode row with Y and flags
- When you are ready, take the short quiz, then continue to carry-select adder